PLATINUM-GROUP METALS

(Palladium, platinum, iridium, osmium, rhodium, and ruthenium)
[Data in kilograms, platinum-group-metal (PGM) content, unless otherwise specified]

Domestic Production and Use: One company in Montana mined and processed PGMs with an estimated value of about $310 million in 2024, a decrease of 42% compared with $541 million in 2023. Small quantities of PGMs also were recovered as byproducts of copper-nickel mining in Michigan; however, this material was sold to foreign companies for refining. The leading domestic use for PGMs was in catalytic converters to decrease harmful emissions from automobiles. PGMs are also used in catalysts for bulk-chemical production and petroleum refining; dental and medical devices; electronic applications, such as in computer hard disks, hybridized integrated circuits, and multilayer ceramic capacitors; glass manufacturing; investment; jewelry; and laboratory equipment.

Salient Statistics—United States:	**2020**	**2021**	**2022**	**2023**	**2024[e]**
Mine production:[1]					
Palladium	14,600	13,700	10,100	10,300	8,000
Platinum	4,200	4,020	3,000	3,040	2,000
Imports for consumption:[2]					
Palladium	76,400	72,600	65,200	66,900	69,000
Platinum	64,900	67,900	64,200	66,800	72,000
PGM waste and scrap	185,000	160,000	41,500	32,000	36,000
Iridium	1,620	2,310	1,610	2,040	1,400
Osmium	1	1	1	—	1
Rhodium	20,700	16,500	13,200	12,100	15,000
Ruthenium	13,900	18,000	13,300	10,800	10,000
Exports:[3]					
Palladium	48,600	43,900	42,200	33,600	39,000
Platinum	28,900	29,400	23,100	11,300	12,000
PGM waste and scrap	33,200	37,800	35,200	13,900	13,000
Rhodium	1,480	1,350	717	453	1,000
Other PGMs	1,440	2,180	1,010	845	2,000
Consumption, apparent:[4,5]					
Palladium	82,300	81,400	74,100	88,600	83,000
Platinum	47,300	51,100	52,900	67,000	71,000
Price, dollars per troy ounce:[6]					
Palladium	2,205.27	2,419.18	2,133.81	1,351.66	980
Platinum	886.02	1,094.31	966.54	973.00	950
Iridium	1,633.51	5,158.40	4,581.93	4,672.78	4,800
Rhodium	11,205.06	20,254.10	15,585.00	6,660.58	4,600
Ruthenium	271.83	576.12	577.02	466.49	440
Employment, mine, number	1,480	1,600	1,560	1,450	900
Net import reliance[5, 7] as a percentage of apparent consumption:					
Palladium	34	35	31	38	36
Platinum	76	75	78	83	85

Recycling: About 120,000 kilograms of palladium and platinum were recovered globally from new and old scrap in 2024, including about 45,000 kilograms of palladium and 8,500 kilograms of platinum recovered from automobile catalytic converters in the United States.

Import Sources (2020–23): Palladium: Russia, 32%; South Africa, 32%; Belgium, 8%, Italy, 8%; and other, 20%. Platinum: South Africa, 45%; Belgium, 12%; Germany, 10%; Italy, 9%; and other, 24%.

Tariff: All unwrought and semimanufactured forms of PGMs are imported duty free. See footnote 2 for specific Harmonized Tariff Schedule of the United States codes.

Depletion Allowance: 22% (domestic), 14% (foreign).

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PLATINUM-GROUP METALS

Government Stockpile:[8]

Material	FY 2024 Potential acquisitions	FY 2024 Potential disposals	FY 2025 Potential acquisitions	FY 2025 Potential disposals
Iridium	—	15	—	15
Platinum	—	261	—	261

Events, Trends, and Issues: Production at a domestic mine continued, but the company placed one of its operations on care-and-maintenance status and reduced the number of its employees owing to the high cost of operations and decreases in PGM prices. Production of PGMs in South Africa, the world's leading producer of PGM-containing mined material, decreased compared with that in 2023 owing to declining prices, higher costs associated with deep-level mining, labor disputes, and ongoing disruptions to the supply of electricity. Production in Russia, the world's leading producer of mined palladium, decreased owing to disruptions from natural disasters, lower metal grades and ore recovery, ongoing issues related to the Russia-Ukraine conflict, and planned outages at a major metallurgical plant.

The estimated annual average price for iridium in 2024 increased by 3% compared with the average price in 2023 whereas the estimated annual average prices for rhodium decreased by 31% for rhodium, by 27% for palladium, by 6% for ruthenium, and slightly for platinum compared with annual average prices in 2023. Price decreases were attributed to decreased demand, investor uncertainty, and oversupply.

World Mine Production and Reserves: Reserves for Russia were revised based on a Government report.

	Mine production				PGM reserves[9]
	Palladium		Platinum		
	2023	2024[e]	2023	2024[e]	
United States	10,300	8,000	3,040	2,000	820,000
Canada	16,100	15,000	5,170	5,200	310,000
Russia[e]	87,000	75,000	21,000	18,000	[10]16,000,000
South Africa	74,900	72,000	125,000	120,000	63,000,000
Zimbabwe	15,900	15,000	19,200	19,000	1,200,000
Other countries	4,200	4,200	5,710	4,600	NA
World total (rounded)	208,000	190,000	179,000	170,000	>81,000,000

World Resources:[9] World resources of PGMs are estimated to total more than 100 million kilograms. The largest resources and reserves are in the Bushveld Complex in South Africa.

Substitutes: Palladium has been used as a substitute for platinum in most gasoline-engine catalytic converters because of the historically lower price for palladium relative to that of platinum. About 25% of palladium can routinely be substituted for platinum in diesel catalytic converters; the proportion can be as much as 50% in some applications. For some industrial end uses, one PGM can substitute for another, but with losses in efficiency.

[e]Estimated. NA Not available. — Zero.

[1]Estimated from published sources.

[2]Includes data for the following Harmonized Tariff Schedule of the United States codes: 7110.11.0010, 7110.11.0020, 7110.11.0050, 7110.19.0000, 7110.21.0000, 7110.29.0000, 7110.31.0000, 7110.39.0000, 7110.41.0010, 7110.41.0020, 7110.41.0030, 7110.49.0010, and 7118.90.0020; 7112.92.0000 (2020–21); and 7112.92.0100 (2022–24).

[3]Includes data for the following Schedule B numbers: 7110.11.0000, 7110.19.0000, 7110.21.0000, 7110.29.0000, 7110.31.0000, 7110.39.0000, 7110.41.0000, and 7110.49.0000; 7112.92.0000 (2020–21); and 7112.92.0100 (2022–24).

[4]Defined as primary production + secondary production + imports – exports.

[5]Excludes imports and (or) exports of waste and scrap.

[6]Engelhard unfabricated metal average annual prices. Source: S&P Global Platts Metals Week.

[7]Defined as imports – exports.

[8]See Appendix B for definitions.

[9]See Appendix C for resource and reserve definitions and information concerning data sources.

[10]Reserves for Russia are based on the Russian Classification system A+B+C1+C2, where C2 are deposits that are being developed or prepared for development.